

ORDER AND ORDER_ITEM MANAGEMENT SYSTEM

Swiggy Instamart E-Commerce Database Management System

1. Database Design

1.1 Orders Table

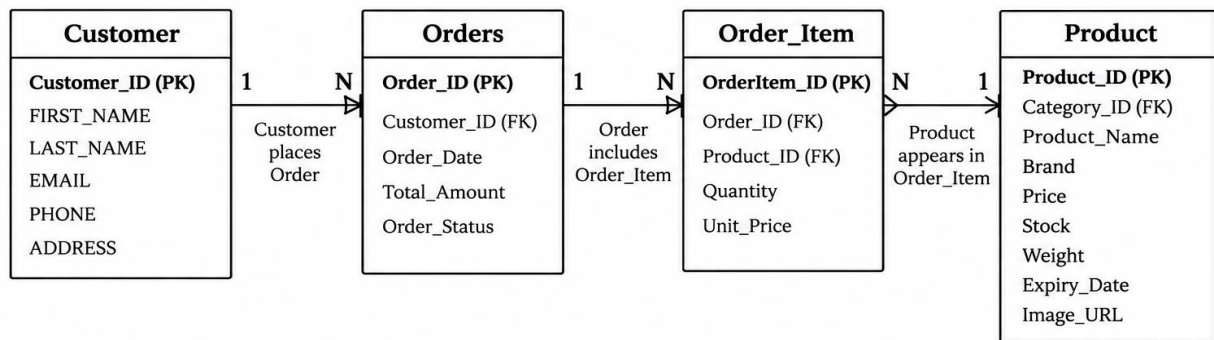
Attribute	Data Type	Key Type / Constraint
Order_ID	NUMBER	Primary Key
Customer_ID	NUMBER	Foreign Key
Order_Date	DATE	NOT NULL
Total_Amount	NUMBER(10,2)	CHECK (> 0)
Order_Status	VARCHAR2(20)	NOT NULL

1.2 Order_Item Table

Attribute	Data Type	Key Type / Constraint
OrderItem_ID	NUMBER	Primary Key
Order_ID	NUMBER	Foreign Key
Product_ID	NUMBER	Foreign Key
Quantity	NUMBER	CHECK (> 0)
Unit_Price	NUMBER(10,2)	CHECK (> 0)

2. Table Relationships

Table Relationships



Foreign Key Connections

Customer.Customer_ID → Orders.Customer_ID
 Orders.Order_ID → Order_Item.Order_ID
 Product.Product_ID → Order_Item.Product_ID

Relationship Summary

Customer → Orders : One-to-Many
 Orders → Order_Item : One-to-Many
 Product → Order_Item : One-to-Many

3. Table Creation

3.1 Create Orders Table

```

SQL> CREATE TABLE Orders (
2     Order_ID NUMBER PRIMARY KEY,
3     Customer_ID NUMBER,
4     Order_Date DATE NOT NULL,
5     Order_Status VARCHAR2(20) NOT NULL,
6     Total_Amount NUMBER(10,2) CHECK (Total_Amount > 0),
7     CONSTRAINT fk_orders_customer
8     FOREIGN KEY (Customer_ID)
9     REFERENCES Customer(Customer_ID)
10 );
    
```

Table created.

3.2 Create Order_Item Table

```

SQL>      CREATE TABLE Order_Item (
2      OrderItem_ID NUMBER PRIMARY KEY,
3      Order_ID NUMBER,
4      Product_ID NUMBER,
5      Quantity NUMBER CHECK (Quantity > 0),
6      Unit_Price NUMBER(10,2) CHECK (Unit_Price > 0),
7      CONSTRAINT fk_orderitem_order
8      FOREIGN KEY (Order_ID)
9      REFERENCES Orders(Order_ID),
10     CONSTRAINT fk_orderitem_product
11     FOREIGN KEY (Product_ID)
12     REFERENCES Product(Product_ID)
13 );

```

Table created.

4. Order Data Insertion

```

SQL> INSERT INTO Orders
2 VALUES (1001, 101, DATE '2026-09-15', 'Placed', 450.00);

1 row created.

SQL> INSERT INTO Orders
2 VALUES (1002, 102, DATE '2026-09-16', 'Delivered', 780.00);

1 row created.

SQL> INSERT INTO Orders
2 VALUES (1003, 103, DATE '2026-09-17', 'Pending', 320.00);

1 row created.

SQL> INSERT INTO Orders
2 VALUES (1004, 104, DATE '2026-09-18', 'Shipped', 650.00);

1 row created.

SQL> INSERT INTO Orders
2 VALUES (1005, 105, DATE '2026-09-19', 'Delivered', 920.00);

1 row created.

SQL> INSERT INTO Orders
2 VALUES (1006, 101, DATE '2026-09-20', 'Placed', 550.00);

1 row created.

SQL> INSERT INTO Orders
2 VALUES (1007, 102, DATE '2026-09-21', 'Delivered', 875.00);

1 row created.

SQL> INSERT INTO Orders
2 VALUES (1008, 103, DATE '2026-09-22', 'Shipped', 430.00);

1 row created.

```

5. Order_Item Data Insertion

```
SQL> INSERT INTO Order_Item VALUES (1, 1001, 1, 2, 150.00);

1 row created.

SQL> INSERT INTO Order_Item VALUES (2, 1001, 2, 1, 120.00);

1 row created.

SQL> INSERT INTO Order_Item VALUES (3, 1002, 3, 3, 80.00);

1 row created.

SQL> INSERT INTO Order_Item VALUES (4, 1002, 4, 2, 200.00);

1 row created.

SQL> INSERT INTO Order_Item VALUES (5, 1003, 5, 1, 320.00);

1 row created.

SQL> INSERT INTO Order_Item VALUES (6, 1004, 1, 2, 150.00);

1 row created.

SQL> INSERT INTO Order_Item VALUES (7, 1005, 2, 4, 120.00);

1 row created.

SQL> INSERT INTO Order_Item VALUES (8, 1006, 3, 2, 80.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Item VALUES (9, 1007, 4, 3, 200.00);

1 row created.

SQL> INSERT INTO Order_Item VALUES (10, 1008, 5, 2, 320.00);
```

6. Order Modification Operations

6.1 Modify Order Amount

```
SQL> SELECT Order_ID, Total_Amount
2  FROM Orders
3  WHERE Order_ID IN (1001, 1002);
```

ORDER_ID	TOTAL_AMOUNT
1001	500
1002	850

6.2 Modify Order Data

```
SQL> UPDATE Orders
2  SET Total_Amount = 500.00,
3      Order_Status = 'Delivered'
4  WHERE Order_ID = 1001;
```

1 row updated.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> UPDATE Orders
2  SET Total_Amount = 850.00,
3      Order_Status = 'Shipped'
4  WHERE Order_ID = 1002;
```

1 row updated.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> SELECT
2  Order_ID,
3  Customer_ID,
4  Order_Date,
5  Total_Amount,
6  Order_Status
7  FROM Orders
8  WHERE Order_ID IN (1001, 1002);
```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS
1001	101	15-SEP-26	500	Delivered
1002	102	16-SEP-26	850	Shipped

7. Order_Item Modification

10 rows selected.

```
SQL> UPDATE Order_Item
  2  SET Quantity = 3,
  3      Unit_Price = 160.00
  4  WHERE OrderItem_ID = 1;
```

1 row updated.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> UPDATE Order_Item
  2  SET Quantity = 2,
  3      Unit_Price = 210.00
  4  WHERE OrderItem_ID = 2;
```

1 row updated.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> SELECT
  2  OrderItem_ID,
  3  Order_ID,
  4  Product_ID,
  5  Quantity,
  6  Unit_Price
  7  FROM Order_Item
  8  WHERE OrderItem_ID IN (1, 2);
```

ORDERITEM_ID	ORDER_ID	PRODUCT_ID	QUANTITY	UNIT_PRICE
1	1001	1	3	160
2	1001	2	2	210

8. Customer Order History Report

```

SQL> SELECT
  2     c.Customer_ID,
  3     c.FIRST_NAME || ' ' || c.LAST_NAME AS Customer_Name,
  4     o.Order_ID,
  5     o.Order_Date,
  6     p.Product_Name,
  7     oi.Quantity,
  8     oi.Unit_Price,
  9     o.Total_Amount,
10     o.Order_Status
11 FROM Customer c
12 JOIN Orders o
13     ON c.Customer_ID = o.Customer_ID
14 JOIN Order_Item oi
15     ON o.Order_ID = oi.Order_ID
16 JOIN Product p
17     ON oi.Product_ID = p.Product_ID
18 WHERE c.Customer_ID = 101
19 ORDER BY o.Order_Date;

```

CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID	ORDER_DATE	PRODUCT_NAME	QUANTITY	UNIT_PRICE	TOTAL_AMOUNT	ORDER_STATUS
101	Rahul Sharma	1001	15-SEP-26					

9. Customer-wise Order Summary

```

SQL> SELECT
2     c.Customer_ID,
3     c.FIRST_NAME || ' ' || c.LAST_NAME AS Customer_Name,
4     COUNT(o.Order_ID) AS Total_Orders,
5     SUM(o.Total_Amount) AS Total_Amount
6 FROM Customer c
7 JOIN Orders o
8     ON c.Customer_ID = o.Customer_ID
9 GROUP BY
10    c.Customer_ID,
11    c.FIRST_NAME,
12    c.LAST_NAME
13 ORDER BY c.Customer_ID;

```

CUSTOMER_ID

CUSTOMER_NAME

TOTAL_ORDERS TOTAL_AMOUNT

101		
Rahul Sharma	2	1050

102		
Priya Kumar	2	1725

CUSTOMER_ID

CUSTOMER_NAME

TOTAL_ORDERS TOTAL_AMOUNT

103		
Arun Ravi	2	750